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Numerical solutions of reaction-diffusion equations: Application to neural and cardiac models

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We describe the implementation of the explicit Euler, Crank-Nicolson, and implicit alternating direction methods for solving partial differential equations and apply these methods to obtain numerical solutions of three excitable-media models used to study neurons and cardiomyocyte dynamics. We discuss the implementation, accuracy, speed, and stability of these numerical methods. © *2016 American Association of Physics Teachers.*
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I. INTRODUCTION

Many of the most beautiful systems in chemistry, physics, biology, and physiology can be described by reaction-diffusion equations.[1–5] For example, the organization and distribution of vegetation-like bushes in arid ecosystems[6,7] and the stripes and spots on fish,[8] snakes,[9] and the skin and fur of mammals[10,11] can be described by the standing waves produced by reaction-diffusion equations.

Reaction-diffusion equations are a set of partial differential equations that describe the time and spatial changes of at least two interacting variables (x_i) that can represent species, densities, or concentrations. The equations can be written as

$$\frac{dx_i}{dt} = f_i(\{x_1, \ldots, x_n\}) + \nabla \cdot (D_i \nabla x_i) \quad (i = 1, \ldots, n), \quad (1)$$

where the left-hand side represents the rate at which the variable x_i changes in time. The function f_i on the right-hand side determines the production and the degradation of x_i as a function of all the other concentrations. The second term on the right-hand side indicates the rate at which x_i diffuses in space as given by the diffusion coefficient D_i, which can be very different for different variables. If $D_i = 0$, there is no diffusion for x_i, and Eq. (1) reduces to an ordinary differential equation.

In addition to standing waves, solutions of reaction-diffusion equations that describe excitable media[12–14] can lead to dynamical waves such as those observed in predator-prey populations,[15] chemical systems,[5] the retina,[16] the heart,[17] and the brain.[18] An excitable medium is a nonlinear dynamical system characterized by a stable resting state, a threshold above which the medium can be excited, and a refractory period during which a second excitation cannot re-excite the system. A forest is a good example of an excitable medium. Normally, the forest is in a stable state (resting state), but when the temperature rises above the ignition point (the threshold), a fire will be generated and propagate through the forest (the medium is excited). A second fire cannot be generated in the burned area until the trees regrow (refractory period).

In the following, we will describe several numerical methods that allow the solution of reaction-diffusion equations in large domains. We will focus on applications to the heart and the brain (excitable media). One motivation for choosing the heart and brain is that large-scale simulations of these systems have shed light on arrhythmic mechanisms and neurological disorders and have aided in the development of new treatments.[19–21] Current experiments using optical mapping allow direct comparisons of many complex patterns observed in tissue with those obtained by simulations.[20] Because the heart and brain are complex nonlinear systems with coupled chemical, electrical, and mechanical phenomena at various biological levels, it is much more difficult to control many factors in experiments compared to numerical solutions. Hence, simulations offer an efficient way of combining all the factors and studying them at various scales.

The propagation of excitations in neurons through synapses and cardiomyocytes through gap junctions can be modeled by reaction-diffusion equations. However, unlike many systems where all species diffuse, only the voltage diffuses while the ionic concentrations are local in cardiac and neuronal tissues. In this case Eq. (1) takes the form

$$C_m \frac{du}{dt} = -I_{\text{ion}}(u, \{v_j\}) + \nabla \cdot (D \nabla u) \quad (2a)$$

$$\frac{dv_i}{dt} = y_i(u, \{v_j\}). \quad (2b)$$

Here, C_m is the membrane capacity, which we choose to be $1\ \mu\text{F/cm}^2$ for simplicity. The transmembrane voltage u is the voltage difference between the intracellular and extracellular

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space. The conductivity coefficient D describes the connections between cells and can be a complicated matrix to account for anisotropy such as fiber direction in the brain and the heart. The total ionic current $I_{\text{ion}}(u,\{v_j\})$ is the current leaking across the membrane surface between the intracellular and extracellular space. The gating variable v_j represents the opening or closing of a particular ion channel j. Expressions for $I_{\text{ion}}(u,\{v_j\})$ and v_j vary greatly.[22] Different relations have been proposed for $I_{\text{ion}}(u,\{v_j\})$ and v_j and are usually named after their authors. For example, Hodgkin-Huxley[23] is a neuronal model based on voltage-clamping experiments on squid giant axons and remains one of the major achievements in the field. Its ionic current is composed of a sodium component I_{Na}, a potassium component I_{K}, and a leaking current I_{ℓ}, with three gating variables, m, h, and n. Its form for the gating variables has been widely adopted by later models, including the Beeler-Rueter model,[24] which was the first mammalian ventricular muscle cell model. This muscle cell model has four ionic currents with six gating variables.[25]

We will also consider the FitzHugh-Nagumo model,[26] which has only one gating variable v (all variables and parameters are dimensionless)

$$\frac{du}{dt} = -I_{\text{ion}}(u,v) + I_{\text{stim}} + D\nabla^2 u, \tag{3a}$$

$$I_{\text{ion}} = u(u-\alpha)(u-1) + v, \tag{3b}$$

$$\frac{dv}{dt} = \epsilon(\beta u - \gamma v - \delta), \tag{3c}$$

with α, β, γ, and δ constants. There are three differences in Eq. (3a) compared to Eq. (2a). We omitted the capacity C_m because it is a constant, and we simplified the problem by assuming the tissue is isotropic; that is, the conductivity coefficient D in Eq. (2a) is a single constant instead of a matrix. We also added an external stimulus current term I_{stim}, which is a pulse-like periodic current source, injected from the left end of the tissue, to pace (stimulate) the cable, imitating a heart beat. The period of the stimulus (the time interval between two sequential stimuli) is called the pacing cycle length.

Figure 1 shows the transmembrane voltage u measured at the center cell on a cable (1D) as a function of time for the FitzHugh-Nagumo model under physiological conditions with constant pacing (I_{stim} is a periodic pulse function). The fast rising and falling of the transmembrane voltage u is called an action potential. During the fast upstroke of the action potential, the ionic current flows into the intracellular space, increasing the transmembrane voltage; this phase is called depolarization. When positive ions exit the intracellular space, the transmembrane voltage drops; this phase is called repolarization. The action potential is initiated by injecting external current into the left end of the cable, which propagates toward the right in the cable through the connections between cells (synapses for neural cells and gap junctions for cardiac cells). This propagation is described by the diffusion term in the reaction-diffusion equation. Excitable cells like neurons and cardiac muscles can conduct cell-to-cell communication through the propagation of the action potential. In the heart, for example, contraction can be coordinated in cardiac muscles by the propagation of action potentials.

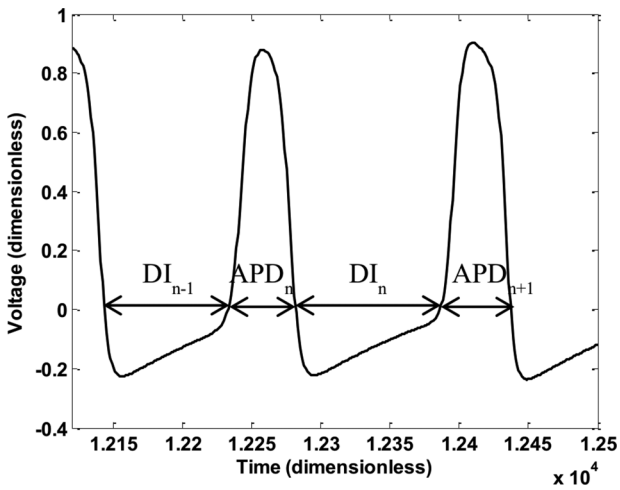


Fig. 1. Illustration of the action potential duration (APD) and the diastolic interval (DI).

The duration of the action potential captures the systolic duration when blood is ejected from the heart chambers. The diastolic interval is the recovery time between two excitations when the heart chambers are refilled. If the action potential duration is not adjusted to a faster heart rate, the diastolic interval will be reduced, leading to insufficient refilling of the blood. In reality, when the heart beats at a higher frequency, the ionic gates are not fully recovered, resulting in a reduced ionic current, and thus a smaller action potential duration and diastolic interval. The action potential duration in cardiac cells is two orders of magnitude longer than in neurons due to the activation of calcium in cardiac cells, which allows contraction and extends the plateau of the action potential.

The restitution curve relates the duration of an excitation to the previous amount of time that the system is unexcited. Excitable systems are said to restitute because to return them to a former condition is time dependent; the function that gives this information is called the restitution function. The plot of the duration of the action potential versus its preceding diastolic interval is known as the action potential duration restitution curve. The preceding diastolic interval affects not only the duration of the action potential but also its propagation speed. The conduction velocity restitution curve shows the wave front propagation velocity at a given site as a function of the proceeding diastolic interval at that site.

The action potential duration restitution and the conduction velocity restitution curves are commonly used in the electrophysiology analysis of cardiac and neuronal models. They can be obtained in experiments using live hearts by varying the period of stimulation.[27,28] They relate the pulse duration and its velocity to the previous dynamical state and can explain the behavior of action potential waves.[29,30]

In addition to plane waves, other solutions of Eq. (3a) in two dimensions (2D) exist, such as spiral waves.[31] The latter are observed in many biological, chemical, and physical systems.[32,33] In a normal heart, the action potentials propagate as an almost simultaneous wave. However, under pathological conditions, an action potential might not complete its normal pathway but rather circle back on itself, forming long-lasting self-sustained excitations that take the form of a spiral wave (in 2D) or a scroll wave (in 3D). Spiral waves in the heart are responsible for deadly arrhythmias such as tachycardias and fibrillations.[17]

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To solve Eq. (3a) for neuronal and cardiac cells and the more general case of Eq. (1), many numerical methods have been developed. We will describe how to implement the explicit Euler, the Crank-Nicolson, and the implicit alternating direction methods.[36,37] We will demonstrate these methods by solving the Hodgkin-Huxley,[23] FitzHugh-Nagumo,[26] and Beeler-Rueter[24] models in isotropic 1D and 2D domains with Neumann boundary conditions and compare their accuracy, stability, and cost. We will also describe how to implement periodic boundary conditions in 1D. Although we will consider these models as examples, the implementation of these methods is model independent.

II. INTEGRATION OF ORDINARY DIFFERENTIAL EQUATIONS FOR SINGLE CELLS

The ordinary differential equation

$$C_m \frac{du}{dt} = -I_{\text{ion}} \tag{4}$$

is known as the zero-dimensional equation. It is a special case of the reaction-diffusion equation without the spatial diffusion term ($D = 0$). Equation (4) describes how the transmembrane voltage behaves in one cell. The ionic current I_{ion} is normally a function of the transmembrane voltage u and gating variables, which are typically also described by ordinary differential equations [see, for example, Eq. (3c)]. There exist many methods for solving such equations, including first-order explicit Euler, second-order implicit Euler, and Runge-Kutta.[36]

We first demonstrate how to use the explicit Euler method to solve Eqs. (3c) and (4). This method is based on expanding the solution in a Taylor series

$$v(t+\Delta t) = v(t) + \left.\frac{dv}{dt}\right|_t \Delta t + \frac{1}{2}\left.\frac{d^2v}{dt^2}\right|_t \Delta t^2 + O(\Delta t^3). \tag{5}$$

To first-order in Δt, the discretized time step, we write the derivative as

$$\frac{dv}{dt} = \frac{v(t+\Delta t) - v(t)}{\Delta t} + O(\Delta t), \tag{6}$$

and substitute into Eq. (3c) to obtain

$$\frac{v(t+\Delta t) - v(t)}{\Delta t} + O(\Delta t) = \epsilon[\beta u(t) - \gamma v(t) - \delta]. \tag{7}$$

Ignoring higher order terms and solving for $v(t+\Delta t)$ gives

$$v(t+\Delta t) = v(t) + \epsilon[\beta u(t) - \gamma v(t) - \delta]\Delta t. \tag{8}$$

Similarly, for Eq. (4) we have

$$u(t+\Delta t) = u(t) - I_{\text{ion}}(t)\Delta t. \tag{9}$$

The explicit Euler method solves an ordinary differential equation by marching forward in time and is also called the forward Euler method. From Eq. (7) we can see that this method is first order in Δt.

There are two main computational costs: the integration in time [Eq. (8) and (9)] and the computation of I_{ion} [or example, Eq. (3b)]. The former can be accelerated by more efficient methods, and the latter can be sped up by using look-up tables. The latter is particularly useful when the equations for I_{ion} involve complicated functions that can increase the execution time when called repetitively. The form of the ionic current for the FitzHugh-Nagumo model is a polynomial of the only gating variable v and voltage u, and therefore, it takes little time to calculate at each time step. In contrast, the Beeler-Rueter model has a much more complicated functional form for the ionic current. For example, the fast Potassium current I_{K_1} has multiple exponentials

$$I_{K_1} = 0.35\left\{\frac{4[e^{0.04(u+85)} - 1]}{e^{0.08(u+53)} + e^{0.04(u+53)}} + \frac{0.2(u+23)}{1 - e^{-0.04(u+23)}}\right\}. \tag{10}$$

The look-up table technique includes two steps. The first step is to compute I_{K_1} in Eq. (10) for a range of u values that are physically possible with a high degree of discretization and save the values in an array before the rest of the calculation starts. The second step is to obtain the values of I_{K_1} from a look-up table (an array) and compute the index of the voltage u in the table and linearly interpolate the values from the table at each time step to obtain an accurate approximation.[25] By using this technique, time-costly functions such as exponentials, hyperbolic, and trigonometric functions are calculated before starting the time loop, and thus do not need to be computed at each time step.

III. INTEGRATION IN ONE-DIMENSIONAL CABLES OR RINGS

To integrate the reaction-diffusion equation for a cable (actually a line because we ignore the thickness of the cable), we need to choose boundary conditions for the ends of the cable. In most cases, a zero-flux (Neumann) boundary condition is used to conserve charge so that no charge is lost at the boundaries. Thus, the model is described as

$$C_m \frac{du}{dt} = -I_{\text{ion}} + D\frac{d^2u}{dx^2} \tag{11a}$$

with

$$D\left.\frac{du}{dx}\right|_{x=0} = D\left.\frac{du}{dx}\right|_{x=L} = 0, \tag{11b}$$

where L is the length of the cable. Another possible experimental situation is a ring where periodic boundary conditions are used instead, so that $u|_{x=0} = u|_{x=L}$. We will use Neumann boundary conditions for most of our discussion.

To integrate Eq. (11), we need to discretize the spatial domain as we did for the time domain. In 1D, the domain can be discretized into $N_x - 1$ elements with N_x nodes, $\Delta x = L/(N_x - 1)$, $x_i = (i-1)L/(N_x - 1)$, $i = 1, 2, \ldots, N_x$. The subscripts denote the grid points and superscripts denote the time steps; for example, u_i^n is the transmembrane potential at the grid point i and time step n.

A. The explicit Euler method

We can use the explicit Euler method to solve the diffusion equation by marching forward in time, determining the values of u_i^{n+1} on every grid point i at time step $n+1$ from

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the values u_i^n at the previous time step. The time derivative in Eq. (11) is treated the same way as in Eq. (6)

$$\frac{du}{dt} = \frac{u_i^{n+1} - u_i^n}{\Delta t} + O(\Delta t). \tag{12}$$

The spatial derivative is also based on a Taylor expansion

$$u(x \pm \Delta x, t) = u(x,t) \pm \left.\frac{du}{dx}\right|_x \Delta x + \frac{1}{2}\left.\frac{d^2u}{dx^2}\right|_x \Delta x^2 \pm \frac{1}{3!}\left.\frac{d^3u}{dx^3}\right|_x \Delta x^3 + O(\Delta x^4). \tag{13}$$

By adding $u(x+\Delta x, t)$ and $u(x-\Delta x, t)$ in Eq. (13), we obtain

$$u(x+\Delta x,t) + u(x-\Delta x,t) = 2u(x,t) + \left.\frac{d^2u}{dx^2}\right|_x \Delta x^2 + O(\Delta x^4), \tag{14}$$

and therefore, we can solve the Laplacian in 1D to second-order in Δx

$$\frac{d^2u}{dx^2} = \frac{u_{i-1}^n - 2u_i^n + u_{i+1}^n}{\Delta x^2} + O(\Delta x^2). \tag{15}$$

We used the Taylor expansion only for the two adjacent grid points $x \pm \Delta x$. If more grid points are used, the Laplacian can be approximated to higher accuracy.

We substitute Eqs. (15) and (12) into Eq. (11a) and obtain

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} + O(\Delta t) = -I_{\text{ion,i}}^n + D\frac{u_{i-1}^n - 2u_i^n + u_{i+1}^n}{\Delta x^2} + O(D\Delta x^2). \tag{16}$$

The above equation is an explicit method that evaluates the diffusion term at time step n so we can compute u_i^{n+1} explicitly in terms of the previous data. The differentiation in time uses a forward method which is a first-order approximation; differentiation in space is a second-order approximation. The total truncation error is $O(\Delta t + D\Delta x^2)$. If we ignore higher order terms, we have

$$u_i^{n+1} = -I_{\text{ion,i}}^n \Delta t + r u_{i-1}^n + (1-2r)u_i^n + r u_{i+1}^n, \tag{17}$$

where $r = D(\Delta t/\Delta x^2)$.

For Neumann boundary conditions, the first derivative of u is zero at the boundary. To keep second-order accuracy in space, we need to use the center difference approximation as in Eq. (15). From a Taylor expansion, we obtain

$$\left.\frac{du}{dx}\right|_{i=1} = \frac{u_2^n - u_0^n}{2\Delta x} + O(\Delta x^2) = 0 \tag{18a}$$

$$\left.\frac{du}{dx}\right|_{i=N_x} = \frac{u_{N_x+1}^n - u_{N_x-1}^n}{2\Delta x} + O(\Delta x^2) = 0. \tag{18b}$$

If we ignore higher order terms, we express the second-order approximation for Neumann boundary conditions as $u_0^n = u_2^n$, $u_{N_x+1}^n = u_{N_x-1}^n$, which compares with $u_0^n = u_1^n$, $u_{N_x+1}^n = u_{N_x}^n$, for which the accuracy is only first-order in Δx. Because the real integration domain is $1 \le i \le N_x$, the points $i=0$ and $i = N_x + 1$ are called ghost cells and are used to account for the boundary conditions. For periodic boundaries (a ring), we have $u_1 = u_{N_x}$.

The truncation error of the explicit Euler method is $O(\Delta t + D\Delta x^2)$. The method is stable[25] only if $r \le \frac{1}{2}$, which means that Δt must decrease at least as $\Delta x^2/D$. Note that for small Δx, the value of Δt must be much smaller.

B. The Crank-Nicolson method

The Crank-Nicolson method is an implicit method and evaluates every term in Eq. (11a) at time step $(n+1/2)$ so that time derivatives are also evaluated using the center difference approximation. The truncation error for Crank-Nicolson is $O(\Delta t^2 + D\Delta x^2)$ and is second-order in both time and space; this method is unconditionally stable[25]

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} + O(\Delta t^2) = -I_{\text{ion,i}}^{n+1/2} + \frac{D}{\Delta x^2}\left(\frac{u_{i-1}^n + u_{i-1}^{n+1}}{2} - 2\frac{u_i^n + u_i^{n+1}}{2} + \frac{u_{i+1}^n + u_{i+1}^{n+1}}{2}\right) + O(D\Delta x^2). \tag{19}$$

We ignore higher-order terms and obtain

$$-\frac{1}{2}r u_{i-1}^{n+1} + (1+r)u_i^{n+1} - \frac{1}{2}r u_{i+1}^{n+1} = -I_{\text{ion,i}}^{n+1/2}\Delta t + \frac{1}{2}r u_{i-1}^n + (1-r)u_i^n + \frac{1}{2}r u_{i+1}^n, \tag{20}$$

where the $n+1$ time step terms are on the left-hand side and the n time step terms are on the right-hand side. The ionic current at time step $(n+1/2)$ can be approximated by

$$I_{\text{ion,i}}^{n+1/2} = I_{\text{ion,i}}^n + \frac{1}{2}\left(I_{\text{ion,i}}^n - I_{\text{ion,i}}^{n-1}\right), \tag{21}$$

an interpolation known as the Adams-Bashforth method.[38] The boundary condition is treated the same way as in the explicit Euler method.

Equation (20) can be written in matrix form as $AU^{n+1} = b^n$, where U and b are both $N_x \times 1$ vectors; A is a $N_x \times N_x$ tridiagonal matrix for the Neumann boundary condition and is a semi-tridiagonal matrix for periodic boundary conditions. It is easy and fast to solve tridiagonal systems without a costly matrix inversion by using the Thomas algorithm.[25]

C. Comparison of the two methods in one dimension

The complexity of the explicit Euler method in one time loop is $O(N_x)$. Crank-Nicolson is an implicit method, requires solving a tridiagonal matrix system, and its complexity is also $O(N_x)$. To obtain the same accuracy, explicit Euler is slower than Crank-Nicolson because the former needs to use a smaller value of Δt than Crank-Nicolson. We can use this criterion as a guideline for choosing values of Δx and Δt as well as for estimating the simulation time. This conclusion also depends on the form of the ionic current, and it is important to always check convergence of the solution.

We will compare the stability and accuracy of the explicit Euler method and the Crank-Nicolson method by comparing

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the action potential shape, the action potential duration restitution curve, the conduction velocity restitution curve, and the maximum conduction velocity of the FitzHugh-Nagumo, Hodgkin-Huxley, and Beeler-Rueter models.

Our protocol for obtaining the restitution curves is to deliver a train of stimuli [injecting external current through the term I_{stim} in Eq. (3)] with the pacing cycle length starting from T_i and decreasing by ΔT every beat of pacing until reaching T_f. The action potential duration, diastolic interval, and conduction velocity are calculated for each beat.

To calculate the action potential duration, diastolic interval, and conduction velocity, we first set up a threshold value u_0 [we use 90% of the voltage repolarization as the cutoff, meaning that $u_0 = u_{\min} + 0.1(u_{\max} - u_{\min})$]. Every time the voltage passes the threshold, we save the time so that the action potential duration, diastolic interval, and conduction velocity can be calculated.[25]

IV. INTEGRATION IN TWO-DIMENSIONAL TISSUE

The 2D reaction-diffusion equation with the Neumann boundary condition is similar to 1D with the Laplacian modified to account for diffusion and the boundary conditions in the y direction

$$\frac{du}{dt} = -I_{\text{ion}} + D\frac{d^2u}{dx^2} + D\frac{d^2u}{dy^2} \tag{22a}$$

with

$$D\frac{du}{dx}\bigg|_{x=0} = D\frac{du}{dx}\bigg|_{x=L_x} = 0 \tag{22b}$$

and

$$D\frac{du}{dy}\bigg|_{y=0} = D\frac{du}{dy}\bigg|_{y=L_y} = 0, \tag{22c}$$

with L_x and L_y the lengths of the domain in the x and the y directions, respectively. The x and y directions are divided, respectively, into $N_x - 1$ and $N_y - 1$ segments, each having N_x and N_y nodes. The total number of nodes is $N = N_xN_y$.

Because the 2D Crank-Nicolson method is simply an extension of the 1D case by discretizing the Laplacian in the y-direction in the same way as in the x-direction, we will not discuss it here.[25] The conclusions about errors and stability are also similar to 1D.[25] However, instead of a tridiagonal matrix system, we need to solve a tridiagonal matrix with two additional bands located at the N_xth sub-diagonal and super-diagonal positions. We show how this change influences the computation time in Sec. VI.

A. The alternating direction implicit method

The alternating direction implicit method is a finite-difference method for solving parabolic, hyperbolic, and elliptic partial differential equations in two or more dimensions. The idea is to simplify the calculation by using only tridiagonal matrices and to solve one direction (x) explicitly while fixing the other direction (y) implicitly, and then reversing the procedure in the next step. In this way, the integration matrix has a tridiagonal form at both steps and simple tridiagonal matrix algorithms can be applied to solve the problem easily. The method in 2D is composed of two $\Delta t/2$ steps.

Step 1: Evaluate the Laplacian in the x-direction at time step $(n + 1/2)$ and evaluate the Laplacian in the y-direction at time step n. The left panel of Fig. 2 shows an example of ordering the grids, starting with the x-direction. We define

$$D_x^+ u_{i,j}^n = u_{i+1,j}^n - u_{i,j}^n, \tag{23a}$$

$$D_x^- u_{i,j}^n = u_{i,j}^n - u_{i-1,j}^n, \tag{23b}$$

$$D_x u_{i,j}^n = u_{i+1,j}^n - u_{i-1,j}^n, \tag{23c}$$

$$D_x^2 u_{i,j}^n = D_x^+ D_x^- u_{i,j}^n = u_{i+1,j}^n - 2u_{i,j}^n + u_{i-1,j}^n. \tag{23d}$$

We use this notation and similar operators for the y-direction to write the first step as

$$\frac{u_{i,j}^{n+1/2} - u_{i,j}^n}{\Delta t/2} = -I_{\text{ion,i,j}}^{n+1/2} + \frac{D}{\Delta x^2}D_x^2 u_{i,j}^{n+1/2} + \frac{D}{\Delta y^2}D_y^2 u_{i,j}^n. \tag{24}$$

We separate the left-hand side with terms at time step $(n + 1/2)$ (unknown) and the right-hand side with terms at time step n (known)

$$\begin{aligned} &-\frac{1}{2}r_x u_{i-1,j}^{n+1/2} + (1 + r_x)u_{i,j}^{n+1/2} - \frac{1}{2}r_x u_{i+1,j}^{n+1/2} \\ &\quad = -\frac{1}{2}I_{\text{ion,i,j}}^{n+1/2}\Delta t + \frac{1}{2}r_y u_{i,j-1}^n + (1 - r_y)u_{i,j}^n + \frac{1}{2}r_y u_{i,j+1}^n, \end{aligned} \tag{25}$$

where $r_x = D(\Delta t/\Delta x^2)$ and $r_y = D(\Delta t/\Delta y^2)$. Note that the value of Δx and Δy can differ.

Step 2: In the second half of Δt, we evaluate the Laplacian in the y-direction at time step $(n + 1)$ and in the x-direction at time step $(n + 1/2)$, ordering grids in the y-direction first (right panel of Fig. 2)

$$\frac{u_{i,j}^{n+1} - u_{i,j}^{n+1/2}}{\Delta t/2} = -I_{\text{ion,i,j}}^{n+1/2} + \frac{D}{\Delta x^2}D_x^2 u_{i,j}^{n+1/2} + \frac{D}{\Delta y^2}D_y^2 u_{i,j}^{n+1}. \tag{26}$$

Again, we separate the left-hand side terms at time step $(n + 1)$ (unknown) and the right-hand side are terms at time step $(n + 1/2)$ (calculated in step 1)

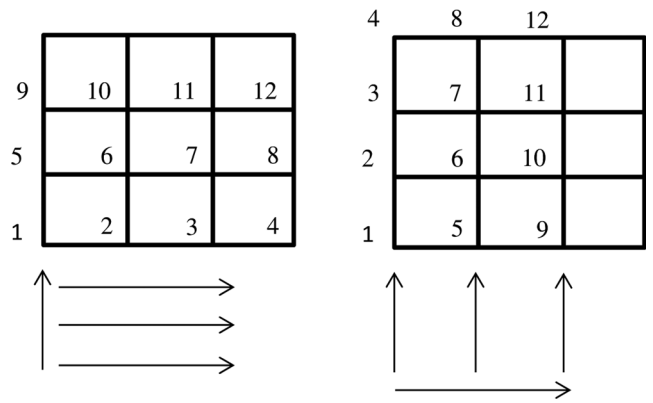


Fig. 2. The order of the grids in the alternating direction implicit method. Step 1 (left): x-direction first; step 2 (right): y-direction first.

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$$-\frac{1}{2}r_y u_{i,j-1}^{n+1} + (1+r_y)u_{i,j}^{n+1} - \frac{1}{2}r_y u_{i,j+1}^{n+1} = -\frac{1}{2}I_{\text{ion,i,j}}^{n+1/2}\Delta t + \frac{1}{2}r_x u_{i-1,j}^{n+1/2} + (1-r_x)u_{i,j}^{n+1/2} + \frac{1}{2}r_x u_{i+1,j}^{n+1/2}. \quad (27)$$

We can write Eqs. (25) and (27) in matrix form as

$$A1U^{n+1/2} = b_1^n \quad (28a)$$

$$A2U^{n+1} = b_2^{n+1/2}. \quad (28b)$$

The tridiagonal matrices $A1$ and $A2$ are provided as the supplementary material.[25] The 2D alternating direction implicit method has local truncation error of $O(\Delta t^2 + D\Delta x^2 + D\Delta y^2)$ and is unconditionally stable in 2D.

B. Comparison of the two methods in two dimensions

The Crank-Nicolson and alternating direction methods are both unconditionally stable and have similar truncation errors, and hence it is expected that their errors have similar asymptotic values for the same values of Δt, Δx, and Δy.

The coefficient matrices on the left-hand side of the alternating direction implicit method are tridiagonal. The most basic method for solving a tridiagonal system is Gauss elimination,[37] which is equivalent to decomposing the matrix into an upper and a lower triangular matrix. In this simple case, it becomes the Thomas algorithm[37] and no fill-ins in the matrix decomposition are generated, meaning that the upper and lower triangular matrices have the same number of nonzero elements as the original coefficient matrix. The coefficient matrices of the Crank-Nicholson method are tridiagonal plus one sub-diagonal band and one super-diagonal band (a banded matrix); the "distance" between the diagonal and the sub-diagonal (or super-diagonal) band is N_x. Using Gauss elimination to solve such a matrix generates additional nonzero elements in the upper and lower triangular matrices within the bandwidth and adds extra complexity. Therefore, solving a linear problem of the form $Ax = b$ with a tridiagonal matrix is much faster than with a banded matrix. The additional cost of the alternating direction implicit method is the requirement of two steps for every Δt, in addition to reordering the grids at every step.

To compare the two methods, we will use both to solve the dynamics of spiral waves in the FitzHugh-Nagumo and Beeler-Rueter models and compare their accuracy by calculating the spiral wave tip trajectories.

A spiral wave is generated through symmetry breaking. We will use the FitzHugh-Nagumo model as an example to demonstrate the symmetry breaking mechanism (see Fig. 3). On a square domain, the wave is initiated on one side of the tissue at $y = 0$ [panel (a)]. When the wave propagates to the

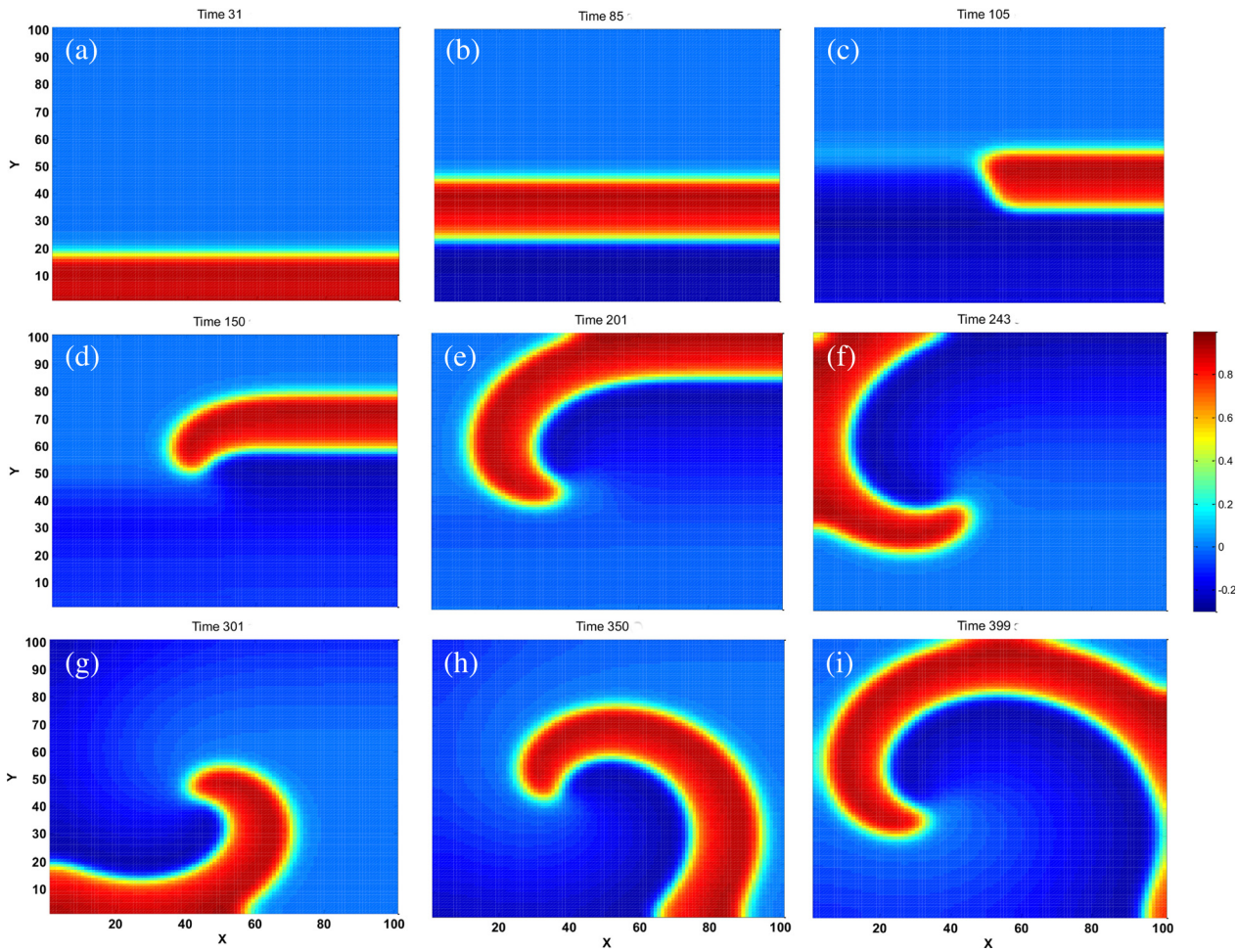


Fig. 3. Spiral wave evolution in the FitzHugh-Nagumo model computed using the alternating direction implicit method. The domain size is 100×100, $\Delta x = \Delta y = 1$, and $\Delta t = 0.2$. The spiral wave is shown for the first 400 time units.

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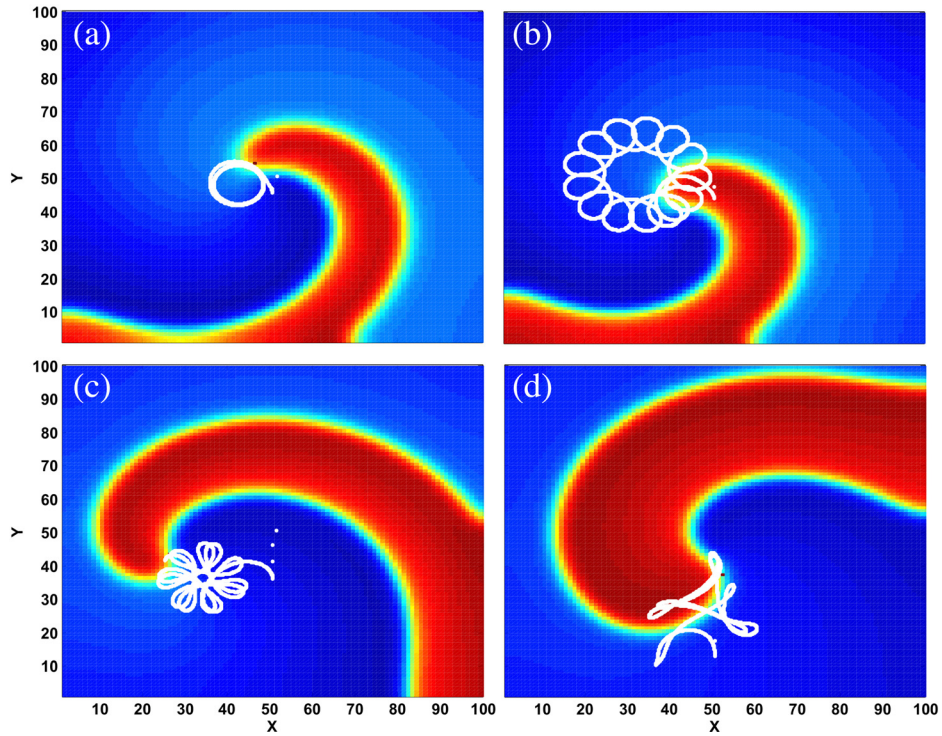


Fig. 4. Different types of spiral tip trajectories produced by changing the parameter ϵ in the FitzHugh-Nagumo model: (a) $\epsilon = 0.012$, (b) $\epsilon = 0.010$, (c) $\epsilon = 0.005$, and (d) $\epsilon = 0.002$. Panel (a) shows a rigid rotation where the spiral shape remains constant and the tip of the spiral traces a circle. Panels (b)–(d) show more complicated flower-like patterns, called meandering spiral waves, with outward-facing petals.

center of the domain [panel (b)], we break symmetry by setting the voltage on the left half of the plane to the resting potential [panel (c)]. This procedure effectively breaks the plane wave as shown in panels (b) and (c), and forces the wave front to meet with the wave back at some point that has no defined phase (it is neither a front nor a back) and can be described as a point with zero normal velocity.[39] This point is now the tip of the spiral wave that rotates [panels (d)–(i)].

A spiral wave is usually characterized by the spiral tip trajectory. There are several methods to identify the spiral wave tip.[34,35] One way is to define the tip as the point in the voltage contour with zero normal velocity. Figure 4 shows four spiral wave tip trajectories generated from the FitzHugh-Nagumo model by changing the value of the parameter ϵ.

To find the tip of the spiral wave, we look for a point on a given isovoltage contour line with zero normal velocity. We first chose a voltage threshold u_{iso} and create isopotential lines in space. The tip (or point with zero normal velocity) can be defined as the intersection of two isopotential lines, one at time t_n and another at t_{n+1}. The tip position $\vec{r} = (x, y)$ satisfies $u^n(\vec{r}) = u^{n+1}(\vec{r}) = u_{\text{iso}}$. Suppose the tip (x, y) is located between four adjacent points on a square (see Fig. 5): (x_i, y_j), (x_{i+1}, y_j), (x_{i+1}, y_{j+1}), and (x_i, y_{j+1}). We denote $\bar{x} = (x - x_i)/\Delta x$ and $\bar{y} = (y - y_j)/\Delta y$, so that $0 \le \bar{x}, \bar{y} \le 1$. We can approximate $u^n(\vec{r})$ and $u^{n+1}(\vec{r})$ with the bilinear interpolation formulas

$$(1-\bar{x})(1-\bar{y})u^n_{i,j} + \bar{x}(1-\bar{y})u^n_{i+1,j} + (1-\bar{x})\bar{y}u^n_{i,j+1} + \bar{x}\bar{y}u^n_{i+1,j+1} = u_{\text{iso}} \tag{29a}$$

$$(1-\bar{x})(1-\bar{y})u^{n+1}_{i,j} + \bar{x}(1-\bar{y})u^{n+1}_{i+1,j} + (1-\bar{x})\bar{y}u^{n+1}_{i,j+1} + \bar{x}\bar{y}u^{n+1}_{i+1,j+1} = u_{\text{iso}}. \tag{29b}$$

There are two unknown variables ($\bar{x}$ and $\bar{y}$) and two quadratic equations, and thus the solution is unique. After solving for $\bar{x}$ and $\bar{y}$, we can obtain the position of the tip using $x = x_i + \bar{x}\Delta x$ and $y = y_j + \bar{y}\Delta y$.[32,39]

V. ONE-DIMENSIONAL RESULTS

We solve the Hodgkin-Huxley, FitzHugh-Nagumo, and Beeler-Rueter models with a stimulus current I_{stim} inserted at the left end of the cable and compare the waves that propagate along the cable. For the Hodgkin-Huxley and Beeler-Rueter models, we chose $L = 3$ cm and $L = 3.5$ cm, respectively. The diffusion coefficient is $D = 1 \times 10^{-3}$ cm^2/ms, which is close to the values obtained from physiological measurements.[23,40] Because the FitzHugh-Nagumo model is phenomenological, we used dimensionless units with $L = 100$ and $D = 1$.

For Hodgkin-Huxley, we tested values of Δx from 0.0125–0.00315 cm and values of Δt from 0.01–0.00125 ms.

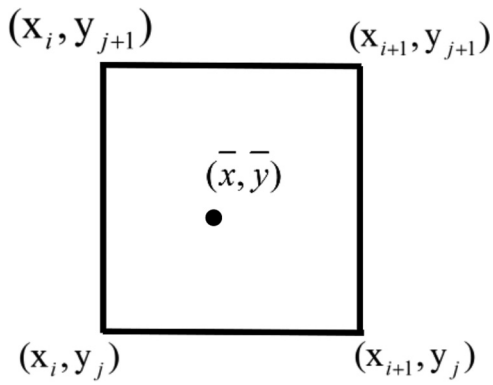


Fig. 5. Bilinear interpolation allows us to find the tip within the grid points.

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For Beeler-Rueter, Δx ranged from 0.025–0.00625 cm and Δt from 0.01–0.0025 ms. We chose smaller values of Δx and Δt for Beeler-Rueter because the dynamics for the sodium current in the Beeler-Rueter model is faster than in the Hodgkin-Huxley model, reflecting the faster physiological upstroke time of cardiac cells compared to neuronal ones. The time scales of the FitzHugh-Nagumo model are similar to the Hodgkin-Huxley model. However, the FitzHugh-Nagumo model is described in generic units, and we used values of Δx ranging from 0.5 to 0.125 and Δt from 0.05 to 0.0125. The factor $r = D\Delta t/\Delta x^2$ that appears in all the integration methods has similar values for the physiological and dimensionless models and varies from $r_{\mathrm{HH}} \in [0.008, 1.01]$ to $r_{\mathrm{FHN}} \in [0.05, 3.2]$.

A. The action potential shape

The shape of the action potential and its changes as a function of the pacing cycle duration are of interest. The action potential in biological cells is generated by a flux of ions. After an activation, many of the ion channels require time to completely recover to their rest states as well as for the ion concentrations to return to their original values. The faster the pacing cycle, the less recovered is a cell, and the more that changes appear in the shape and duration of an action potential. In Fig. 6 we show how the action potential measured in one cell at the middle of the cable changes as a function of pacing period, the integration method, and the values of Δx and Δt for the FitzHugh-Nagumo model. Similar figures for the Hodgkin-Huxley and Beeler-Rueter model are provided as the supplementary material.[25]

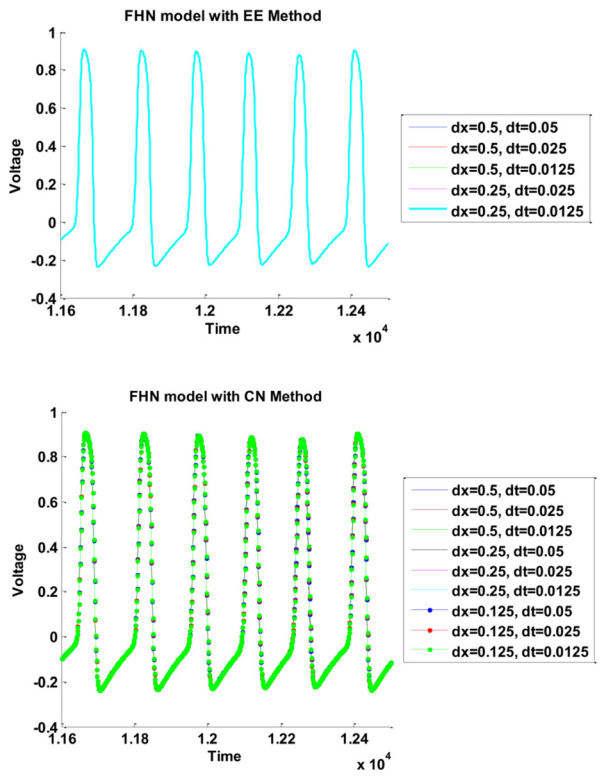


Fig. 6. The action potential of the FitzHugh-Nagumo model for different Δt and Δx values (top: explicit Euler; bottom: Crank-Nicolson). A train of stimuli is delivered with the pacing cycle length starting from 500, decreasing by 10 every beat until it reaches 50. The last ten stimuli are shown, which generate only six excitations due to conduction block. (They fail to produce a propagation because the cells are still in their refractory periods.) The resting potential is zero and the voltage is dropped below the resting potential after depolarization.

As described in Sec. III C, one end of a cable is originally paced at a period T, and after each activation, another excitation is produced at period $T - \Delta T$. In this way, each action potential that propagates along the cable is produced sooner and sooner with a decreasing time of ΔT. From Fig. 6 and Ref. 25, we can make several interesting observations.

For the Hodgkin-Huxley and FitzHugh-Nagumo models, the action potentials are composed of a fast upstroke and a fast downstroke lasting only a few milliseconds (or unit times for FitzHugh-Nagumo), which is characteristic of neuron firing. In contrast, the Beeler-Rueter model has a fast upstroke but a much slower downstroke. Cardiac cells have two distinct time constants (about an order of magnitude difference), an activation time of ≈5 ms and a deactivation time of ≈300–400 ms. Cardiac cells, unlike neurons, have a large calcium component that balances the potassium current, generating the long plateau of the action potential and is responsible for the cell contraction.

For the Hodgkin-Huxley and FitzHugh-Nagumo models, the shape of the action potential and its duration changes much less than that in the Beeler-Rueter model as the pacing cycle length of the stimulation decreases. Cardiac models can develop complex dynamics because the action potential duration is a sensitive function of the diastolic interval at fast pacing. A small change in the diastolic interval will cause avalanches in the action potential duration, as described in Sec. V B.

For the Hodgkin-Huxley and FitzHugh-Nagumo models, the shape of the action potential is almost independent of the integration method or the discretization. In contrast, the integration method and discretization for Beeler-Rueter can affect the solution.

B. The action potential duration restitution and conduction velocity restitution curves

As shown in Sec. V A, changes in the period of stimulation can result in changes in the shape and duration of the action potential. These changes can be seen in the action potential duration restitution and the conduction velocity restitution curves. Figures 7 and 8 show these curves for the FitzHugh-Nagumo model; similar figures for Hodgkin-Huxley and Beeler-Rueter are provided as the supplementary material.[25] The action potential duration and conduction velocity increase to a plateau as a function of the diastolic interval for all three models, but the models exhibit different ranges for the action potential duration, diastolic interval, and conduction velocity.

The major difference between the numerical methods is the convergence rates for the restitution curves. For the FitzHugh-Nagumo model, the action potential duration converges as fast as the conduction velocity; namely, if the action potential duration-diastolic interval converges, the conduction velocity-diastolic interval also converges for the values of Δt and Δx considered (see Figs. 7 and 8). The conduction velocity for Beeler-Rueter (Hodgkin-Huxley) converges much slower (faster) than the action potential duration, meaning that the Δx and Δt values that enable the action potential duration-diastolic interval (conduction velocity-diastolic interval) restitution curves to converge are not small enough for the conduction velocity (action

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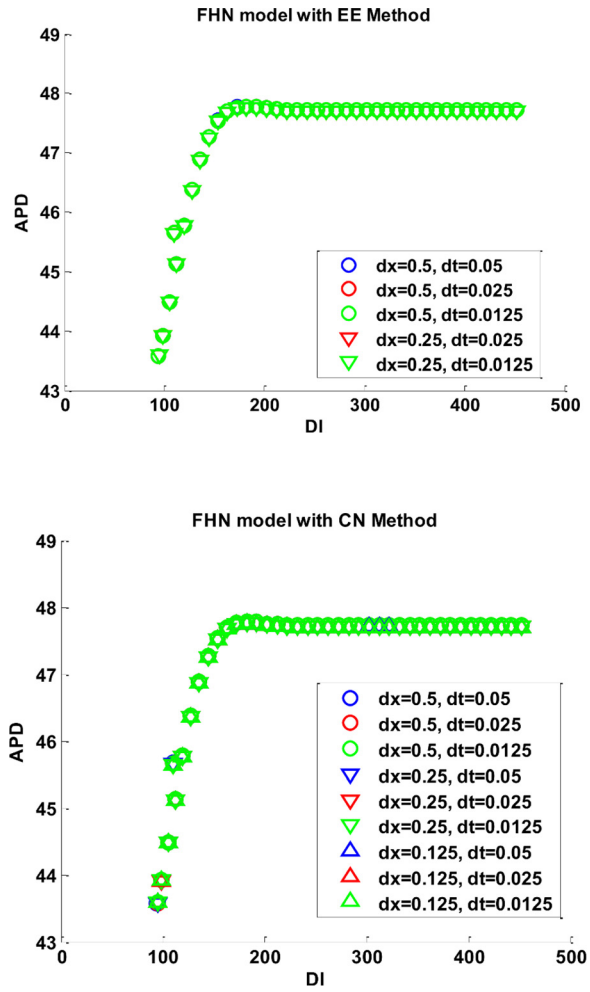


Fig. 7. There is no significant difference of the action potential duration restitution curves of the FitzHugh-Nagumo model obtained by two methods (top: explicit Euler; bottom: Crank-Nicolson) with different Δt and Δx values.

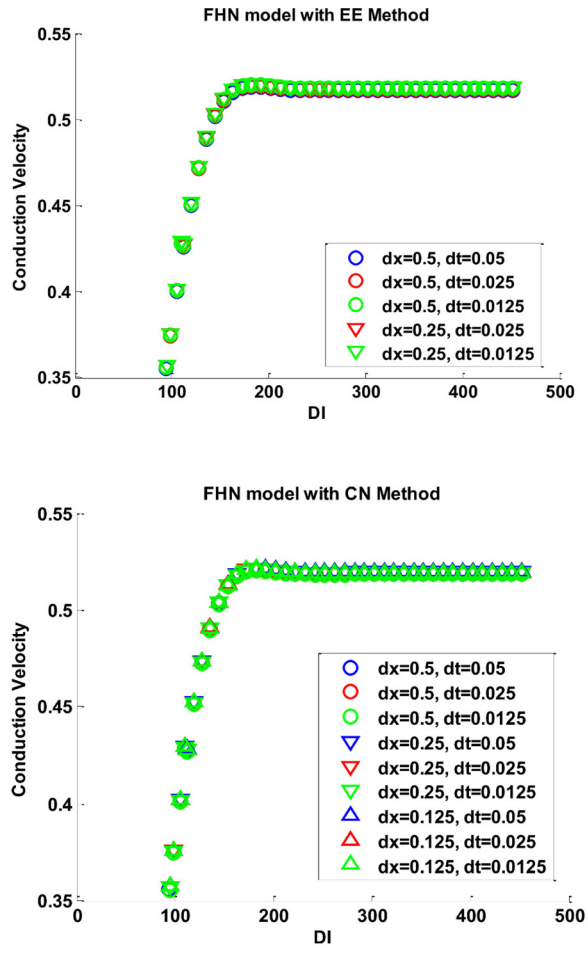


Fig. 8. There is no difference in the conduction velocity restitution curves for the FitzHugh-Nagumo model for the Δt and Δx values considered (top: explicit Euler; bottom: Crank-Nicolson).

potential duration) restitution curves to converge for the Beeler-Rueter (Hodgkin-Huxley) model.

C. The maximum conduction velocity

To further examine the convergence of the models, we focus on the maximum conduction velocity, which is defined as the asymptotic value of the velocity as the diastolic interval goes to infinity. We compute the maximum of the conduction velocity by using the velocity of a stimulus initiated on a cable where all the cells are at rest so that it corresponds to a completely recovered system with an infinite diastolic interval.

The plots in Fig. 9 show the maximum of the conduction velocity plotted against Δt and Δx for the FitzHugh-Nagumo model. Similar figures for the Hodgkin-Huxley and Beeler-Rueter models are provided as the supplementary material.[25] In agreement with the analytical analysis in Sec. III, Crank-Nicolson has the same order of accuracy in space as explicit Euler, but is more accurate in time for large Δx and Δt values. Crank-Nicolson and explicit Euler converge to the same result as $\Delta t \to 0$.

We also observe that for the same Δt and large Δx values, the Crank-Nicolson and explicit Euler methods underestimate the solutions for all three models. The comparisons of increasing/decreasing speeds for each model and each method are summarized in Table I. The trends depend on the model and on the method. For example, for the FitzHugh-Nagumo model, for fixed Δx and decreasing Δt, the maximum conduction velocity increases for the explicit Euler method and decreases for the Crank-Nicolson method, both linearly.

VI. TWO-DIMENSIONAL RESULTS

We demonstrate the evolution of spiral waves and use the tip trajectory as a measure of the accuracy and speed of the two methods. The FitzHugh-Nagumo and the Beeler-Rueter models are used to demonstrate cardiac arrhythmias.

A. The spiral wave

Figure 3 shows the generation and evolution of a spiral wave in the FitzHugh-Nagumo model obtained using the alternating direction implicit method. The spiral wave is generated by symmetry breaking, as described in Sec. IV B.

Panels (a)–(f) in Fig. 10 show the evolution of a spiral wave generated in the Beeler-Rueter model with speed-up calcium dynamics[25,41] obtained by the alternating direction implicit method. The spiral wave has already performed four rotations in only 500 ms, illustrating why spiral waves are so dangerous when occurring in cardiac tissue because they

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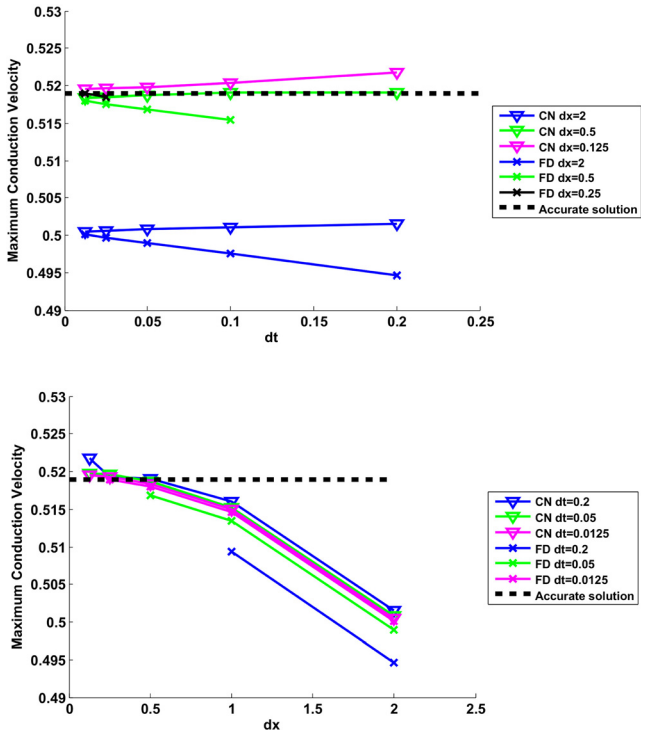


Fig. 9. The plateau (maximum) values of the conduction velocity restitution curves of the FitzHugh-Nagumo model obtained by two methods (top: explicit Euler; bottom: Crank-Nicolson) for different Δt and Δx values.

become sources of extremely fast (and irregular) heart activations.

The dynamics of spiral waves can be complex and depend on both the cell physiology[32,39,41] and/or the tissue structure.[42–44] For the FitzHugh-Nagumo model, spiral waves are stable (as shown in Fig. 3) for most parameters[45] with the exceptions of complex meandering behavior or no spiral formation for some values. There are no instabilities that can lead to spiral wave breakups and chaos, meaning that instead of remaining a spiral shape, the spiral wave "breaks" into multiple wavelets. For the Beeler-Rueter model, there exists stable and highly meandering regimes as well as breakup regimes.[39,41] Panels (g)–(i) in Fig. 10 show spiral wave breakup using the original Beeler-Rueter parameters instead of speed-up calcium as in panels (a)–(f).

The mechanism for breakup in the Beeler-Rueter model is related to T-wave alternans.[46,47] This phenomenon is produced by a period doubling bifurcation and has been explained using the action potential duration and the conduction velocity restitution curve.[29,30] When the slope of the action potential duration restitution curve is greater than one, abrupt changes in the pulse duration can affect the relation between recovery and excitation, modifying the speed of the wave-back (repolarization part of the wave, opposing the wavefront). Once the diastolic interval is in a regime where the action potential duration restitution slope is greater than one, the front and back velocities begin to separate, resulting in a slower wave traveling back compared with the wave traveling forward. The slowly back-propagating wave elongates the refractory period. In a cable, the slow back-propagating wave interacts with the forward-propagating wave of the following beat and blocks its propagation. In 2D the tip rotates toward the slow back wave, encounters the refractory tissues and causes conduction blocks and wave breaks as shown in panels (g)–(i) of Fig. 10

B. The tip trajectory of spiral waves

As mentioned, spiral wave tip trajectories can be used as a measure of the accuracy of a solution. Figure 11 shows the tip trajectories of spirals obtained using the Crank-Nicolson and alternating direction implicit methods for the FitzHugh-Nagumo model. The tip trajectory forms a flower pattern with petals facing outward (a hypocycloidal trajectory). Other parameters can lead to inward petals (epicycloidal) or more complex meandering.[45] Because the convergence is the same for both methods, they produce almost identical trajectories. The tip trajectory for the Beeler-Rueter model has a more complicated pattern[25] due to the complex dynamics discussed in Sec. VI A.

C. The CPU time

Figure 12 shows the CPU time for the two integration methods for the FitzHugh-Nagumo model. We used $\Delta x = \Delta y = 1$ and $\Delta t = 0.2$, as in Fig. 3. The total simulation time is 1000 time units so that the total number of time steps is $1000/0.2 = 5000$; the grid number is $N = N_x N_y$. For example, the grid number $N = 0.9 \times 10^5$ corresponds to a domain size of 300×300. The smallest grid size is 50×50 and the largest is 400×400. The supplementary material[25] provides the CPU time used to generate spiral waves for the Beeler-Rueter model.

Table II shows the fitting parameters obtained from a fit of the rescaled CPU time Y to the rescaled grid number X with $Y = P_1X^2 + P_2X + P_3$, where X and Y are rescaled variables and x and y are the actual grid number and CPU time. For example, $Y = y/(5.8 \times 10^3)$ and $X = x/(1.6 \times 10^5)$ for the FitzHugh-Nagumo model. The values of P_3 are much less than P_1 and P_2, which means when the grid number is zero, the CPU time is also zero. We see that $|P_2| > P_1$ except for the Crank-Nicolson solution of the FitzHugh-Nagumo model

Table I. The convergence of the maximum conduction velocity. For the same Δt and large Δx, explicit Euler (EE) and Crank-Nicolson (CN) underestimate the solution for all three models. For the same Δt, the maximum conduction velocity asymptotically increases for Crank-Nicolson and explicit Euler as Δx decreases for all three models. For the same Δx, the maximum conduction velocity decreases linearly for both methods for Hodgkin-Huxley and Beeler-Rueter as Δt decreases. For FitzHugh-Nagumo, the maximum conduction velocity linearly for the explicit Euler method and decreases linearly for Crank-Nicolson.

		FitzHugh-Nagumo	Beeler-Rueter	Hodgkin-Huxley
Δx fixed, $\Delta t \downarrow$	Explicit Euler	$\uparrow$ linear	$\downarrow$ linear	$\downarrow$ linear
	Crank-Nicolson	$\downarrow$ linear	$\downarrow$ linear	$\downarrow$ linear
	speed	EE > CN	CN > EE	EE > CN
Δt fixed, $\Delta x \downarrow$	Explicit Euler	$\uparrow$ asymptotic	$\uparrow$ asymptotic	$\uparrow$ asymptotic
	Crank-Nicolson	$\uparrow$ asymptotic	$\uparrow$ asymptotic	$\uparrow$ asymptotic

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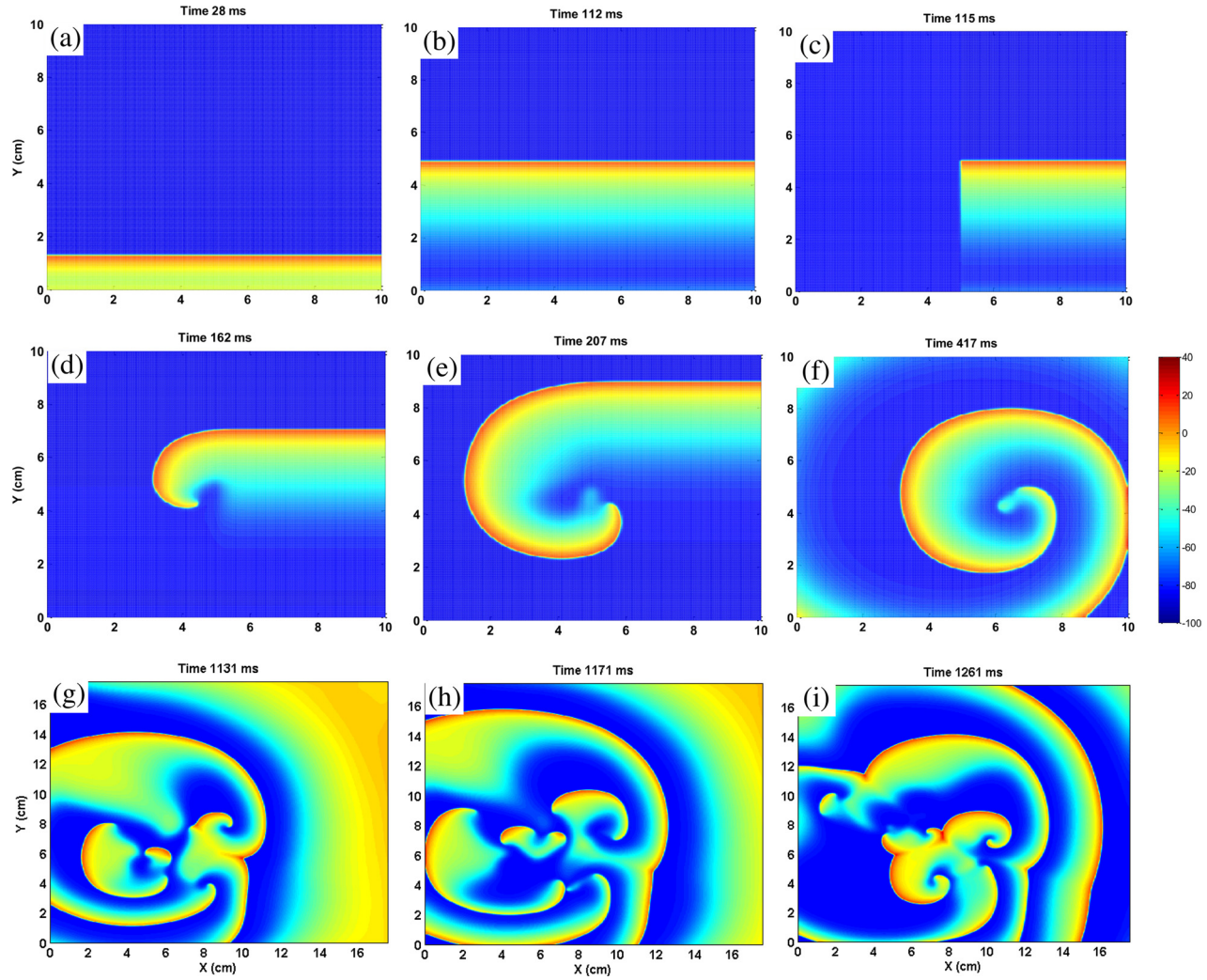


Fig. 10. Spiral wave evolution in the Beeler-Rueter model with speed-up calcium dynamics, calculated using the alternating direction implicit method: (a) initiation; (b) propagation; (c) symmetry break; (d) rotation begins; (e)–(f) spiral wave propagation. The domain size is 10 cm × 10 cm, $\Delta x = \Delta y = 0.025$ cm, $\Delta t = 0.01$ ms. In panels (a)–(f), the spiral wave is shown for the first 500 ms; the times for each frame are indicated in the figure. Panels (g)–(i) show spiral wave breakup using the original parameters of the Beeler-Rueter model. Both the original and speed-up calcium parameters are given in Ref. 25. The domain size is 17.5 cm × 17.5 cm.

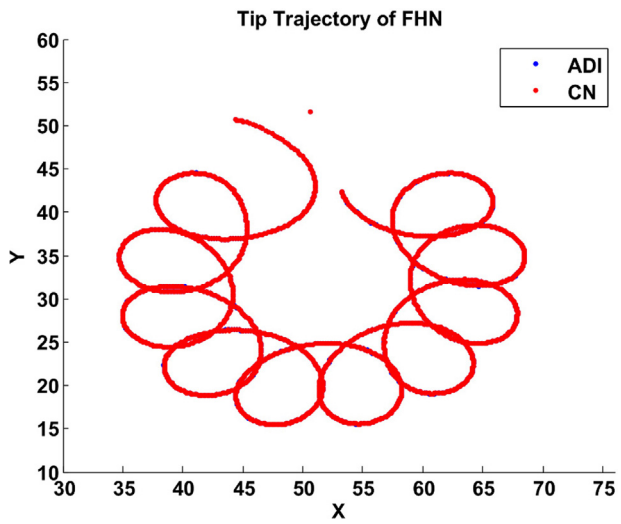


Fig. 11. The tip trajectory of the FitzHugh-Nagumo model using the same domain size, Δt, and Δx as in Fig. 3. The total time to generate this spiral is 2500 time units. The alternating direction implicit and the Crank-Nicolson methods produce almost identical trajectories.

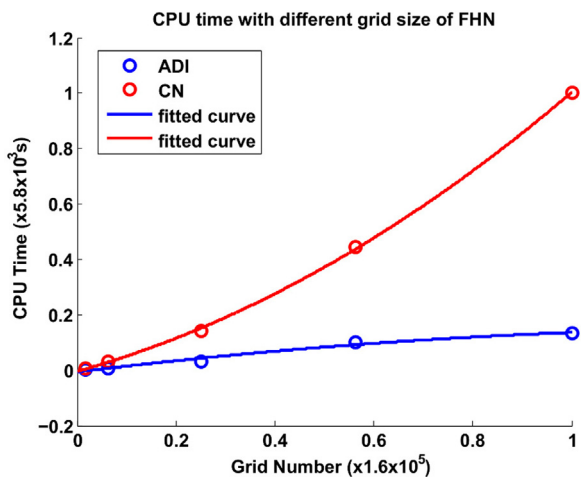


Fig. 12. The CPU time versus grid number for the FitzHugh-Nagumo model with $\Delta t = 0.2$ and $\Delta x = \Delta y = 1$; the total time is 1000.

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Table II. The parameters for the CPU time Y versus the grid number X for different numerical solutions of the FitzHugh-Nagumo and Beeler-Rueter models. The fitting equation is $Y = P_1X^2 + P_2X + P_3$, with $Y = y/(5.8 \times 10^3)$ and $X = x/(1.6 \times 10^5)$ for FitzHugh-Nagumo and $Y = y/(3.7 \times 10^4)$ and $X = x/(1.6 \times 10^5)$ for Beeler-Rueter. Here, y is the CPU time in seconds while X and Y are rescaled so that they are in the range of $[0, 1]$.

Model	Method	P_1	P_2	P_3
FitzHugh-Nagumo	Alternating direction implicit	−0.07338	0.2159	−0.006004
	Crank-Nicolson	0.5095	0.4967	−0.004591
Beeler-Rueter	Alternating direction implicit	0.04088	0.1943	0.0009309
	Crank-Nicolson	0.2939	0.7025	−0.001516

for which $P_1 \approx P_2$, indicating that the CPU time for the two models increases with grid number almost linearly for most methods. The larger value of P_2 for Crank-Nicolson implies that the latter is slower than the alternating direction implicit method, as predicted in Sec. IV B.

These results suggest that for the same Δt and Δx, the Crank-Nicolson and alternating direction implicit methods have similar accuracy, but Crank-Nicolson is much slower than the alternating direction implicit method, especially for large domain sizes. The results are in agreement with our analysis in Sec. IV.

VII. CONCLUSIONS

We discussed three numerical methods for solving reaction-diffusion equations: explicit Euler, Crank-Nicolson, and alternating direction implicit. We applied these methods to solving three excitable media models: FitzHugh-Nagumo (phenomenological), Beeler-Rueter (cardiac), and Hodgkin-Huxley (neural) in one and two dimensions.

In the one-dimensional simulations, equivalent to fibers of cardiac or neuronal tissues, we demonstrated the propagation of action potential waves, and reproduced characteristic curves that are commonly used in experiments to investigate cell dynamics and wave instabilities. We showed how the accuracy of the curves depends on the method and on the choice of Δt and Δx. We found that for the same values of Δt and large Δx, the Crank-Nicolson and explicit Euler methods underestimate the solutions.

We followed the evolution of spiral waves in the FitzHugh-Nagumo and Beeler-Rueter models, and obtained the tip trajectories and the CPU time as a function of domain size. We showed that Crank-Nicolson and the alternating direction implicit have similar accuracy and Crank-Nicolson is much slower than the alternating direction implicit method, especially for the large domain sizes.

We hope this article will be useful for students and researchers who want to study the complex dynamics of the heart and brain, for which simulations can help solve many unanswered questions and elucidate new mechanisms.

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[1]A. M. Turing, "The chemical basis of morphogenesis," Philos. Trans. R. Soc. London B **237**, 37–72 (1952).

[2]*Waves and Patterns Chemical and Biological Media*, edited by H. L. Swinney and V. I. Krinsky (MIT Press, Cambridge, MA, 1991).

[3]P. Ball and N. R. Borley, *The Self-Made Tapestry: Pattern Formation in Nature* (Oxford U.P., Oxford, 1999).

[4]J. E. Pearson, "Complex patterns in a simple system," Science **261**, 189–192 (1993).

[5]*Chemical Waves and Patterns*, edited by R. Kapral and K. Showalter (Kluwer, Dordrecht, 1995), pp. 485–516.

[6]P. Couteron and O. Lejeune, "Periodic spotted patterns in semiarid vegetation explained by a propagation inhibition model," J. Ecol. **89**, 616–628 (2001).

[7]M. Rietkerk, M. C. Boerlijst, F. van Langevelde, R. HilleRisLambers, J. van de Koppel, L. Kumar, and A. M. de Roos, "Self-organization of vegetation in arid ecosystems," Am. Naturalist **160**, 524–530 (2002).

[8]S. Kondo and R. Asai, "A reaction-diffusion wave on the skin of the marine angelfish Pomacanthus," Nature **376**, 765–768 (1995).

[9]J. D. Murray, "A pre-pattern formation mechanism for animal coat markings," J. Theor. Biol. **88**, 161–199 (1981).

[10]S. Kondo, "How animals get their skin patterns: Fish pigment pattern as a live Turing wave," *Systems Biology* (Springer, Japan, 2009), pp. 37–46.

[11]J. D. Murray and M. R. Myerscough, "Pigmentation pattern formation on snakes," J. Theor. Biol. **149**, 339–360 (1991).

[12]E. Meron, "Pattern formation in excitable media," Phys. Rep. **218**, 1–66 (1992).

[13]J. J. Tyson and J. P. Keener, "Singular perturbation theory of traveling waves in excitable media (a review)," Physica D **32**, 327–361 (1988).

[14]A. Pullan, M. Buist, and L. Cheng, *Mathematically Modelling the Electrical Activity of the Heart* (World Scientific, Singapore, 2005), pp. 154–166.

[15]R. Arditi and L. R. Ginzburg, "Coupling in predator-prey dynamics: ratio-dependence," J. Theor. Biol. **139**, 311–326 (1989).

[16]R. O. Wong, "Retinal waves and visual system development," Annu. Rev. Neurosci. **22**, 29–47 (1999).

[17]E. M. Cherry and F. H. Fenton, "Visualization of spiral and scroll waves in simulated and experimental cardiac tissue," New J. Phys. **10**, 125016-1–43 (2008).

[18]A. Hutt, M. Bestehorn, and T. Wennekers, "Pattern formation in intracortical neuronal fields," Comput. Neural Syst. **14**, 351–368 (2003).

[19]S. Luther, F. H. Fenton, B. G. Kornreich, A. Squires, P. Bittihn, D. Hornung, and E. M. Cherry, "Low-energy control of electrical turbulence in the heart," Nature **475**, 235–239 (2011).

[20]F. H. Fenton, S. Luther, E. M. Cherry, N. F. Otani, V. Krinsky, A., Pumir, and R. F. Gilmour, "Termination of atrial fibrillation using pulsed low-energy far-field stimulation," Circulation **120**, 467–476 (2009).

[21]W. Li, A. H. Janardhan, V. V. Fedorov, Q. Sha, R. B. Schuessler, and I. R. Efimov, "Low-energy multistage atrial defibrillation therapy terminates atrial fibrillation with less energy than a single shock," Circ. Arrhythm. Electrophysiol. **4**, 917–925 (2011).

[22]F. H. Fenton and E. M. Cherry, "Models of cardiac cell," Scholarpedia **3**, 1868 (2008).

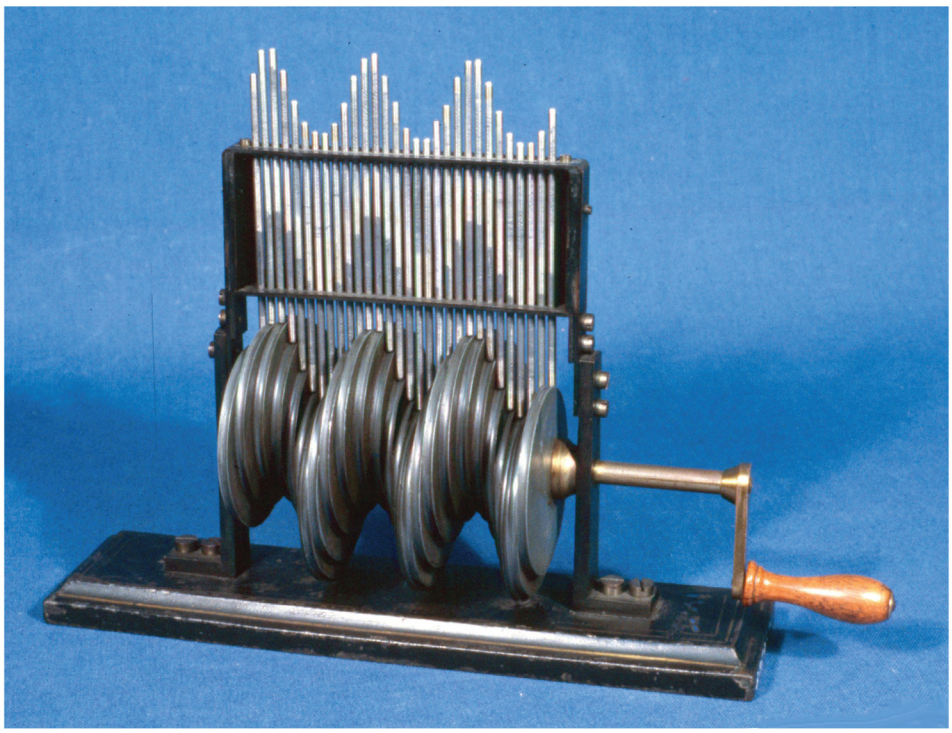
[23]A. L. Hodgkin and A. F. Huxley, "A quantitative description of membrane current and its application to conduction and excitation in nerve," J. Physiol. **117**, 500–544 (1952).

[24]G. W. Beeler and H. Reuter, "Reconstruction of the action potential of ventricular myocardial fibres," J. Physiol. **268**(1), 177–210 (1977).

[25]See Supplementary material at http://dx.doi.org/10.1119/1.4953167 supplies the equations of the ionic models we have discussed, the error associated with the Crank-Nicolson and the alternating direction implicit methods, a derivation of the stability and the matrices for all three finite difference methods, more details of the Thomas algorithm, a discussion of the look-up-table method, and the algorithms to compute the action potential duration, diastolic interval, and conduction velocity. The supplementary material also includes figures for the action potential, action potential

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duration restitution and conduction velocity restitution curves, and maximum conduction velocity of the Hodgkin-Huxley and Beeler-Rueter models in one dimension and figures for the spiral wave tip trajectory and CPU time of the 2D Beeler-Rueter model.
[26]R. FitzHugh, "Mathematical models of threshold phenomena in the nerve membrane," Bull. Math. Biophys. **17**, 257–278 (1955).
[27]V. Elharrar and B. Surawicz, "Cycle length effect on restitution of action potential duration in dog cardiac fibers," Am. J. Physiol. **244**(6), H782–H792 (1983); available at http://www.ncbi.nlm.nih.gov/pubmed/6859281.
[28]M. R. Franz, C. D. Swerdlow, L. B. Liem, and J. Schaefer, "Cycle length dependence of human action potential duration in vivo," J. Clin. Invest. **82**(3), 972–979 (1988).
[29]J. B. Nolasco and R. W. Dahlen, "A graphic method for the study of alternation in cardiac action potentials," J. Appl. Physiol. **25**(2), 191–196 (1968); available at http://www.ncbi.nlm.nih.gov/pubmed/5666097.
[30]M. R. Guevara, G. Ward, A. Shrier, and L. Glass, "Electrical alternans and period doubling bifurcations," Computers in cardiology **562**, 167–170 (1984); available at http://www.medicine.mcgill.ca/physio/guevaralab/Guevara-et-al%201984.pdf.
[31]D. Barkley, M. Kness, and L. Tuckerman, "Spiral-wave dynamics in a simple model of excitable media: The transition from simple to compound rotation," Phys. Rev. A **42**, 2489–2492 (1990).
[32]F. Fenton, E. Cherry, H. Hastings, and S. Evans, "Multiple mechanisms of spiral wave breakup in a model of cardiac electrical activity," Chaos **12**, 852–892 (2002).
[33]S. Jakubith, H. Rotermund, W. Engel, A. von Oertzen, and G. Ertl, "Spatiotemporal concentration patterns in a surface reaction: Propagating and standing waves, rotating spirals, and turbulence," Phys. Rev. Lett. **65**, 3013–3016 (1990).
[34]V. Biktashev, A. Holden, and H. Zhang, "Tension of organizing filaments of scroll waves," Philos. Trans. R. Soc. London A **347**, 611–630 (1994).
[35]A. Panfilov and P. Hogeweg, "Scroll breakup in a three-dimensional excitable medium," Phys. Rev. E **53**, 1740–1743 (1996).
[36]Randal J. LeVeque, *Finite Difference Methods for Ordinary and Partial Differential Equations* (SIAM, Philadelphia, PA, 2007), pp. 13–35, 120–121, 124–130, 181–200.
[37]Yousef Saad, *Iterative Methods for Sparse Linear Systems*, 2nd ed. (SIAM, Philadelphia, PA, 2003), pp. 124–126.
[38]John C. Butcher, *Numerical Methods for Ordinary Differential Equations* (John Wiley & Sons, Hoboken, NJ, 2003).
[39]F. Fenton and A. Karma, "Vortex dynamics in three-dimensional continuous myocardium with fiber rotation: Filament instability and fibrillation," Chaos **8**(1), 20–47 (1998).
[40]A. L. Bueno-Orovio, E. M. Cherry, and F. H. Fenton, "Minimal model for human ventricular action potentials in tissue," J. Theor. Biol. **253**(3), 544–560 (2008).
[41]M. Courtemanche, "Complex spiral wave dynamics in a spatially distributed ionic model of cardiac electrical activity," Chaos **6**(4), 579–600 (1996).
[42]G. Bub, A. Shrier, and L. Glass, "Global organization of dynamics in oscillatory heterogeneous excitable media," Phys. Rev. Lett. **94**, 028105-1–4 (2005).
[43]G. Bub, A. Shrier, and L. Glass, "Spiral wave generation in heterogeneous excitable media," Phys. Rev. Lett. **88**, 058101-1–4 (2002).
[44]E. M. Cherry, J. R. Ehrlich, S. Nattel, and F. H. Fenton, "Pulmonary vein reentry–properties and size matter: Insights from a computational analysis," Heart Rhythm **4**(12), 1553–1562 (2007).
[45]A. T. Winfree, "Varieties of spiral wave behavior: An experimentalist's approach to the theory of excitable media," Chaos **1**, 303–334 (1991).
[46]J. M. Pastore, S. D. Girouard, K. R. Laurita, F. G. Akar, and D. S. Rosenbaum, "Mechanism linking T-wave alternans to the genesis of cardiac fibrillation," Circulation **99**(10), 1385–1394 (1999).
[47]M. A. Watanabe, F. H. Fenton, S. J. Evans, H. M. Hastings, and A. Karma, "Mechanisms for discordant alternans," J. Cardiovasc. Electrophysiol. **12**(2), 196–206 (2001).



Transverse Wave Machine

This little wave machine is designed to show the properties of transverse travelling waves to large audiences – by shadow projection. Despite its obvious appeal, I have seen only three examples in my travels; this one is at the collection of the National Museum of American History at the Smithsonian Institution. In 1900 you could buy it from Max Kohl of Chemnitz, Germany, for 46 marks, or about $11. (Notes and picture by Thomas B. Greenslade, Jr., Kenyon College)

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